Sk Mahiduzzaman



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Experience

University College Cork (UCC), Ireland **Jun 2025 – Sep 2025**
Research Internship — Malware Detection and Mitigation using DICOM images
CNN-based classification was used to process and analyze DICOM medical images. This included developing automated restoration workflows to recover corrupted medical images and simulating malware-infected images for threat detection experiments. Explainable AI (XAI) techniques were also used to interpret model predictions and enhance system transparency and trust.

Education

Adamas University **2022–2026**
B.Tech in Computer Science and Engineering (AI & ML) — **CGPA: 8.28**

P.P.D. High School **2022**
Higher Secondary (10+2) — **92.60%**

P.P.D. High School **2020**
Secondary (10th) — **73.86%**

Skills

Programming Languages : C, Python

Databases : MySQL, MongoDB

AI/ML & GenAI : Machine Learning, Deep Learning, NLP, RAG, LLMs, TensorFlow, PyTorch, LangChain 0.3

Development & Data Tools : FastAPI, Power BI

Development Tools & MLOps : Git, GitHub, Docker

Projects

Book Recommender System (Machine Learning) — Built a recommendation model based on content using TF-IDF vectorization and cosine similarity. Created efficient preprocessing and operations for sparse matrices. Developed a Python interface to generate top book recommendations. GitHub Link

Multimodal Indic Language Identification System (Speech + Text) — Developed a deep learning-based language identification system for 12 Indian languages using speech and text data. Trained and evaluated Transformer-based models including Wav2Vec2, HuBERT, WavLM, mBERT, and XLM-RoBERTa. Implemented data preprocessing, augmentation, feature extraction, model evaluation, and comparative analysis using PyTorch, TensorFlow, and Hugging Face Transformers on an HPC environment. GitHub Link

Automated MLOps Pipeline — Implemented a reproducible ML pipeline using DVC for data versioning. Used Docker for environment consistency and AWS for cloud deployment. Automated training, tracked artifacts, and set up CI/CD workflows for scalable model lifecycle management. GitHub Link

Languages

English · Hindi · Bengali